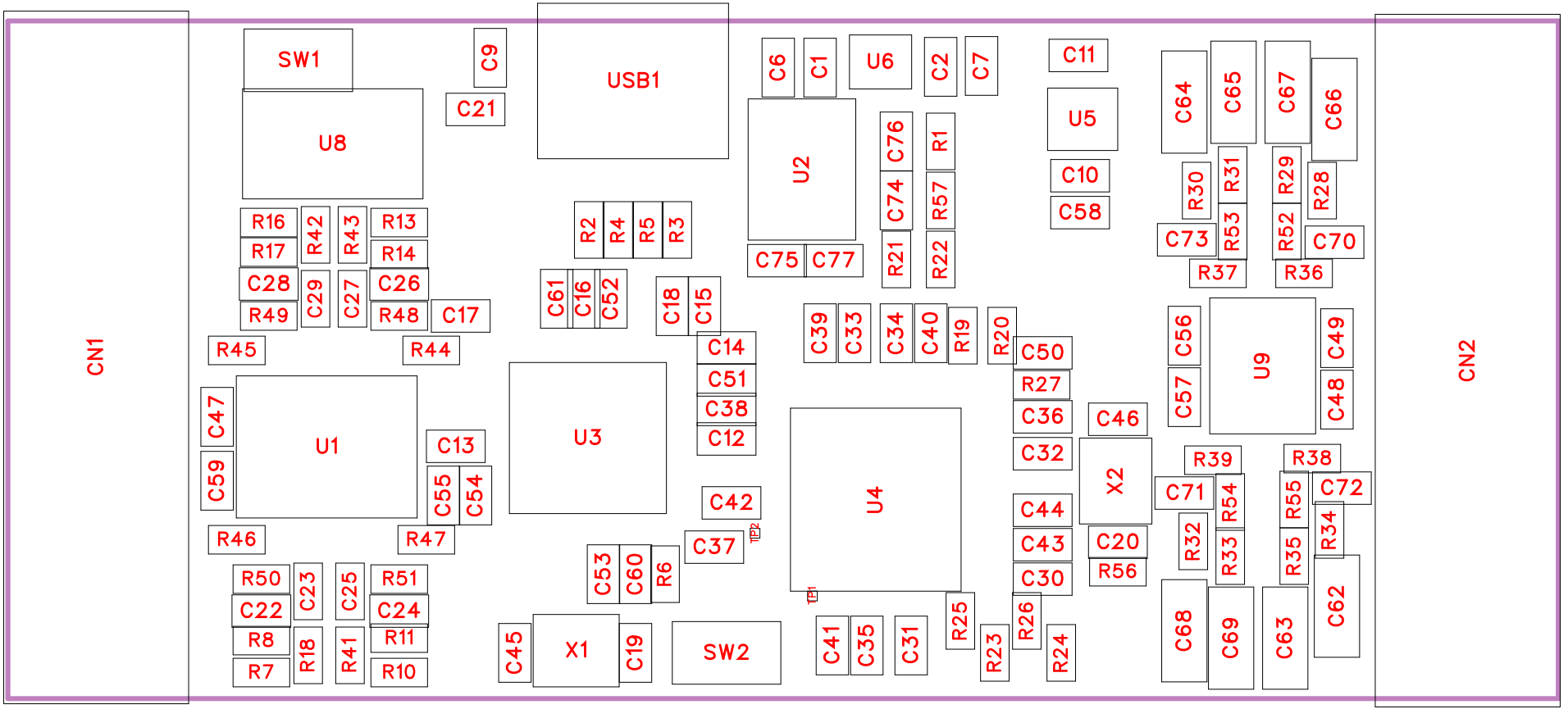
CN1
CN2
SW1
C9
C21
USB1
U8
C6
C1
U6
C2
C7
C11
U5
U2
C76
R1
C74
R57
C10
C58
C64
C65
C67
C66
R30
R31
R29
R28
R52
R53
C73
C70
R37
R36
R16
R42
R43
R13
R17
R14
C28
C29
C27
C26
R49
R48
C17
R45
R44
R2
R4
R5
R3
C75
C77
R21
R22
C61
C16
C52
C18
C15
C39
C33
C34
C40
R19
R20
C56
U9
C49
C57
C48
C14
C51
C38
C12
C50
R27
C36
C32
C46
X2
R39
R38
C71
R54
R55
C72
R32
R33
R35
R34
C20
R56
C44
C43
C30
C47
U1
C59
C13
C55
C54
U3
C42
U4
C37
R46
R47
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C25
R51
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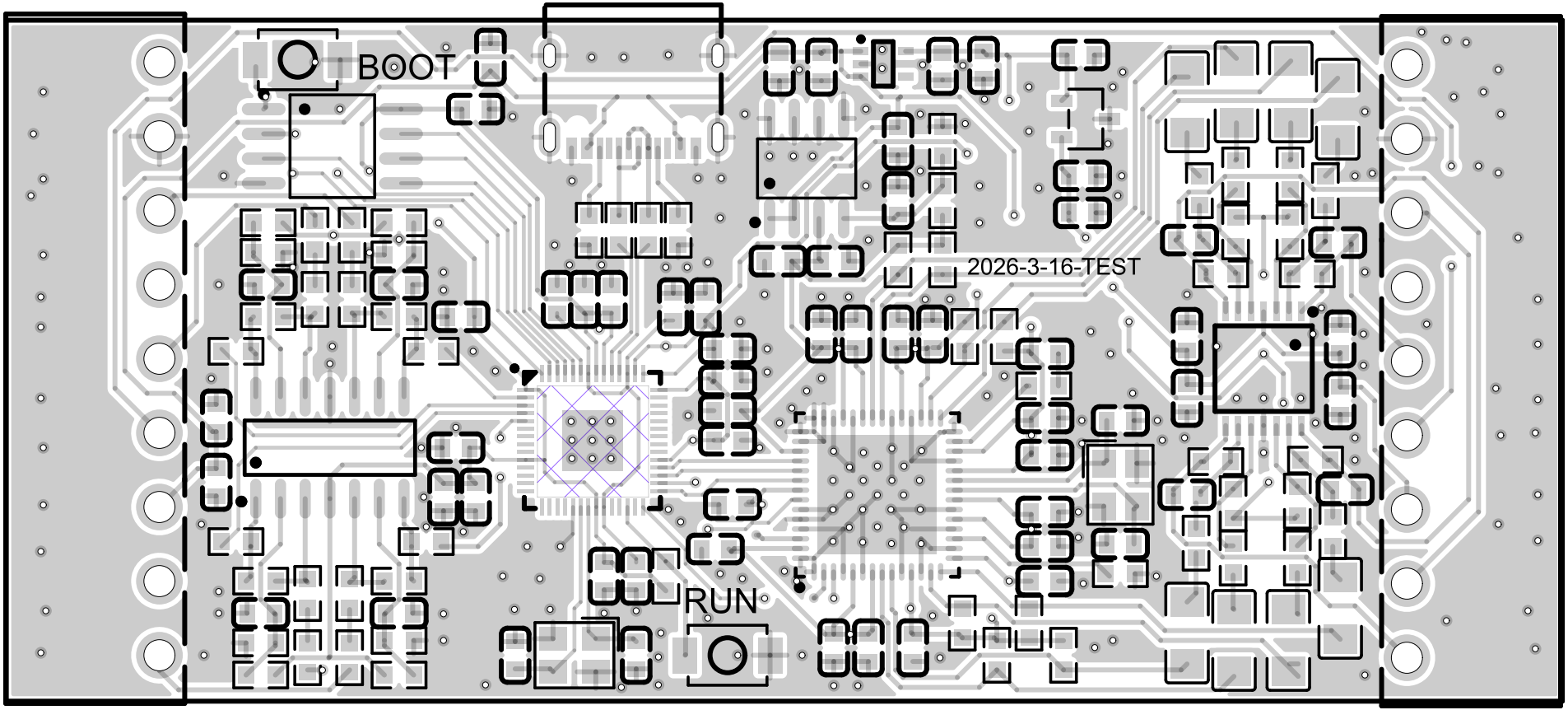
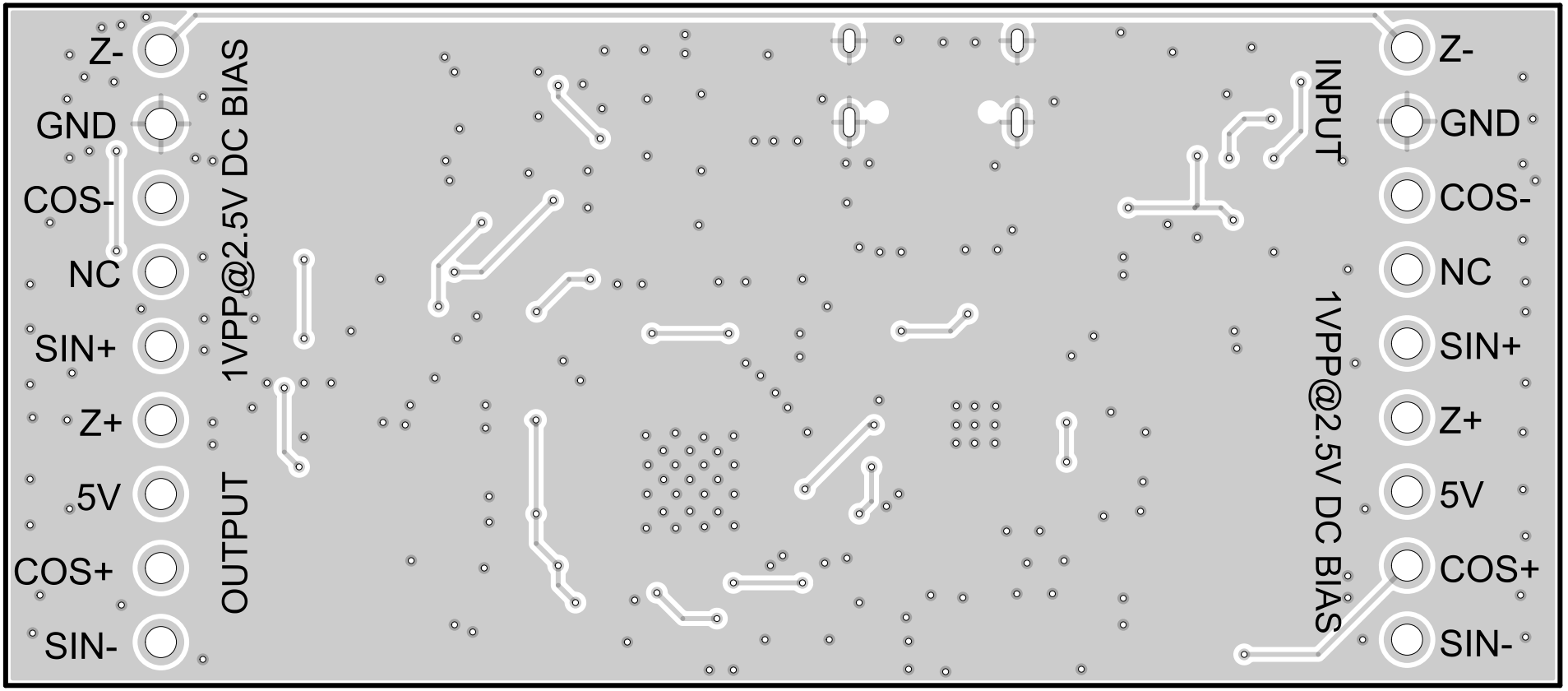
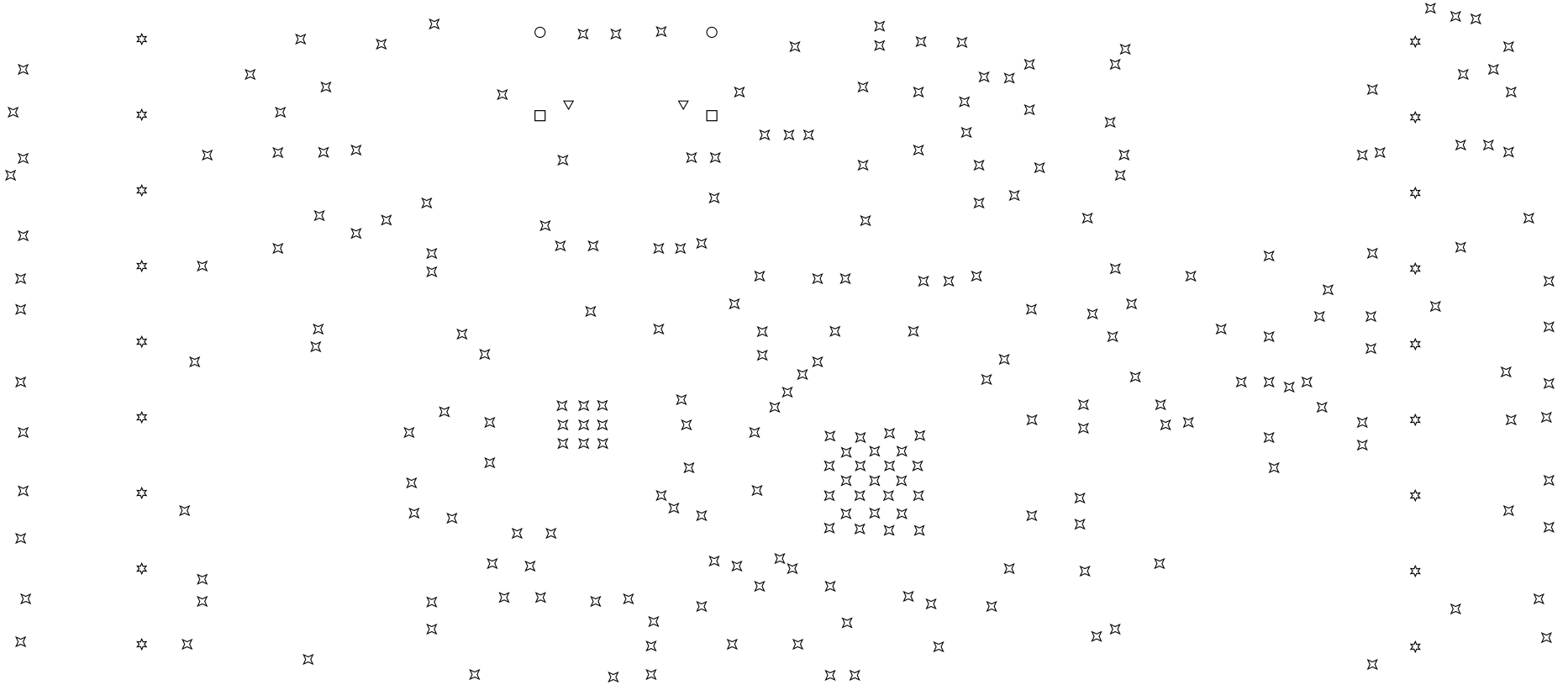
C24
R8
R18
R41
R11
R7
R10
C45
X1
C19
SW2
C41
C35
C31
R25
R23
R26
R24
C68
C69
C63
C62

No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier
1	11	1uF	C1,C2,C10,C11,C18,C22,C24,C26,C28,C54,C56	C0603	1uF	CGA0603X5R105K500JT	HRE(芯声)	C6119852	LCSC
2	6	10uF	C6,C9,C47,C48,C58,C76	C0603	10uF	CL10A106MA8NRNC	SAMSUNG(三星)	C96446	LCSC
3	33	100nF	C7,C12,C13,C14,C15,C16,C17,C21,C30,C31,C32,C33,C34,C35,C36,C37,C38,C39,C40,C41,C42,C43,C44,C49,C51,C52,C53,C55,C57,C59,C60,C61,C77	C0603	100nF	CGA0603X7R104K500JT	HRE(芯声)	C6119867	LCSC
4	4	15pF	C19,C20,C45,C46	C0603	15pF	CL10C150JB8NNNC	SAMSUNG(三星)	C1644	LCSC
5	4	0Ω	C23,C25,C27,C29	R0603	0Ω	RC0603FR-070RL	YAGEO(国巨)	C100044	LCSC
6	1	680pF	C50	C0603	680pF	CC0603JRNPO9BN681	YAGEO(国巨)	C107055	LCSC
7	8	47uF	C62,C63,C64,C65,C66,C67,C68,C69	C1206	47uF	HGC1206R5476M250NSPJ	Chinocera(华瓷)	C22367835	LCSC
8	4	5pF	C70,C71,C72,C73	C0603	5pF	CC0603BRNPO9BN5R0	YAGEO(国巨)	C519109	LCSC
9	1	470pF	C74	C0603	470pF	CL10B471KB8NNNC	SAMSUNG(三星)	C22396	LCSC
10	1	2.2uF	C75	C0603	2.2uF	CL10A225KO8NNNC	SAMSUNG(三星)	C23630	LCSC
11	2	KF2EDGR-3.81-9P	CN1,CN2	CONN-TH_9P-P3.81_KF2EDGR-3.81-9P		KF2EDGR-3.81-9P	KEFA(科发)	C441189	LCSC
12	6	10kΩ	R1,R6,R48,R49,R50,R51	R0603	10kΩ	0603WAF1002T5E	UNI-ROYAL(厚声)	C25804	LCSC
13	2	5.1kΩ	R2,R3	R0603	5.1kΩ	0603WAF5101T5E	UNI-ROYAL(厚声)	C23186	LCSC
14	2	27Ω	R4,R5	R0603	27Ω	RC0603FR-0727RL	YAGEO(国巨)	C137753	LCSC
15	4	24kΩ	R7,R10,R13,R16	R0603	24kΩ	0603WAF2402T5E	UNI-ROYAL(厚声)	C23352	LCSC
16	4	75kΩ	R8,R11,R14,R17	R0603	75kΩ	0603WAF7502T5E	UNI-ROYAL(厚声)	C23242	LCSC
17	12	1kΩ	R18,R28,R29,R30,R31,R32,R33,R34,R35,R41,R42,R43	R0603	1kΩ	0603WAF1001T5E	UNI-ROYAL(厚声)	C21190	LCSC
18	8	50Ω	R19,R20,R21,R22,R23,R24,R25,R26	R0603	50Ω	RT0603BRE0750RL	YAGEO(国巨)	C849792	LCSC
19	1	0Ω	R27	R0603	0Ω	0603WAF0000T5E	UNI-ROYAL(厚声)	C21189	LCSC
20	4	2.4kΩ	R36,R37,R38,R39	R0603	2.4kΩ	0603WAF2401T5E	UNI-ROYAL(厚声)	C22940	LCSC

21	4	33Ω	R44,R45,R46,R47	R0603	33Ω	0603WAF330JT5E	UNI-ROYAL(厚声)	C23140	LCSC
22	4	2kΩ	R52,R53,R54,R55	R0603	2kΩ	0603WAF2001T5E	UNI-ROYAL(厚声)	C22975	LCSC
23	1	3.92kΩ	R56	R0603	3.92kΩ	RT0603BRC073K92L	YAGEO(国巨)	C705716	LCSC
24	1	22kΩ	R57	R0603	22kΩ	0603WAF2202T5E	UNI-ROYAL(厚声)	C31850	LCSC
25	2	TS-1088-AR02016	SW1,SW2	SW-SMD_L3.9-W3.0-P4.45		TS-1088-AR02016	XUNPU(讯普)	C720477	LCSC
26	2	Test-Point	TP1,TP2	Test-Point-0.5mm					
27	1	MCP6564-E/SL	U1	SOIC-14_L8.7-W3.9-P1.27-LS6.0-BL		MCP6564-E/SL	MICROCHIP(美国微芯)	C640925	LCSC
28	1	MIC5209YM-TR	U2	SOIC-8_L4.9-W3.9-P1.27-LS6.0-BL		MIC5209YM-TR	MICROCHIP(美国微芯)	C461593	LCSC
29	1	RP2040	U3	LQFN-56_L7.0-W7.0-P0.4-EP		RP2040	Raspberry Pi(树莓派)	C2040	LCSC
30	1	AD9959BCPZ-REEL7	U4	LFCSP-56_L8.0-W8.0-P0.50-BL-EP		AD9959BCPZ-REEL7	ADI(亚德诺)	C653947	LCSC
31	1	REF3025AIDBZR	U5	SOT-23-3_L2.9-W1.6-P1.90-LS2.8-BR		REF3025AIDBZR	TI(德州仪器)	C11334	LCSC
32	1	ME6211C33R5G	U6	SOT-353_L2.1-W1.3-P0.65-LS2.3-BL		ME6211C33R5G	MICRONE(南京微盟)	C235316	LCSC
33	1	W25Q32JVSSIQ	U8	SOIC-8_L5.3-W5.3-P1.27-LS8.0-BL		W25Q32JVSSIQ	Winbond(华邦)	C179173	LCSC
34	1	OPA4354AIPWR	U9	TSSOP-14_L5.0-W4.4-P0.65-LS6.5-BL		OPA4354AIPWR	TI(德州仪器)	C527494	LCSC
35	1	TYPE-C 16PIN 2MD(073)	USB1	USB-C-SMD_TYPE-C-6PIN-2MD-073		TYPE-C 16PIN 2MD(073)	SHOU HAN(首韩)	C2765186	LCSC
36	1	12MHz	X1	CRYSTAL-SMD_4P-L3.2-W2.5-BL	12MHz	YC12MKBCD2XT	YJX(雅晶鑫)	C52125128	LCSC
37	1	25MHz	X2	CRYSTAL-SMD_4P-L3.2-W2.5-BL	25MHz	YC25MDBCD2XT	YJX(雅晶鑫)	C52164305	LCSC


BOOT
2026-3-16-TEST
RUN


Z-
GND
COS-
NC
SIN+
Z+
5V
COS+
SIN-
1VPP@2.5V DC BIAS
OUTPUT
INPUT
1VPP@2.5V DC BIAS
Z-
GND
COS-
NC
SIN+
Z+
5V
COS+
SIN-